



EAST WEST UNIVERSITY

Physics Lab
Department of MPS

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Name of the Experiment : Determination of spring constant
and effective mass of a given spiral spring.

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Experiment No: 01

Determination of spring constant and effective mass of a given spiral spring.

$$T = 2\pi \sqrt{\frac{m+m'}{k}} = 2\pi \sqrt{\frac{M}{k}} \dots \textcircled{1}$$

$m' =$ effective mass of the spring

$$k = \text{spring constant} = F/x = mg/x$$

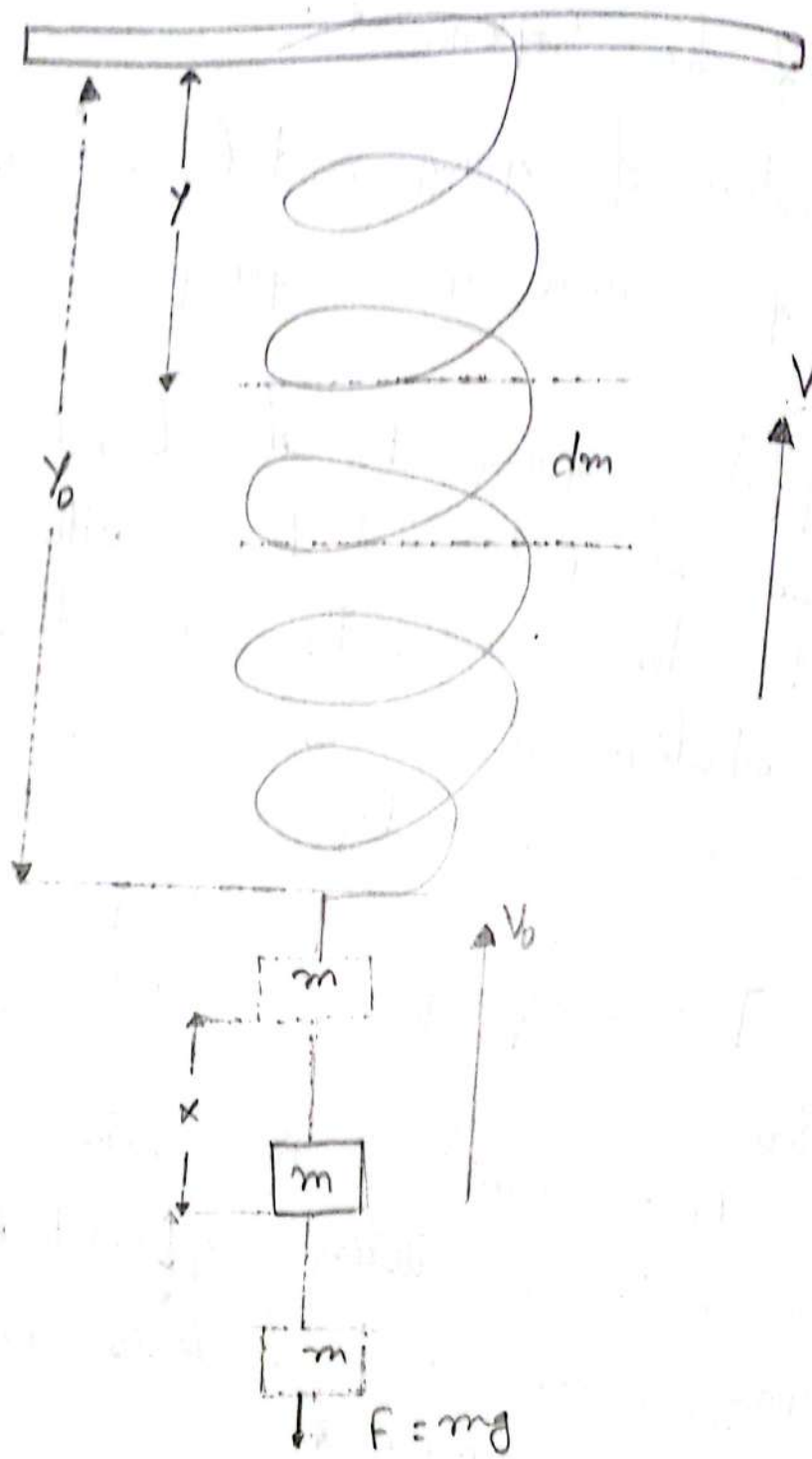


Figure - 1

How the mass of the spring contributes to the effective mass of the vibrating system (load and spring) can be shown as follows.

Consider the kinetic energy of a spring and its load undergoing simple harmonic oscillation. At the instant under consideration let the load m be moving upwards with ~~vel~~ velocity v_0 as shown in the figure.

At this same instant an element of dm of the total mass m_0 of the spring will also be moving up but with a velocity v which is less than v_0 .

It is evident that the ratio between v and v_0 is just the ratio between y and y_0 . (in figure - 1)

Hence $\frac{v}{v_0} = \frac{y}{y_0}$

So, $v = \frac{y \times v_0}{y_0}$ (2)

kinetic energy of small part of mass dm of the spring will be $= \frac{1}{2} dm v^2$

Therefore kinetic energy of full part of the spring along will be $= \int_0^m \frac{1}{2} v^2 dm$

From similarity we can write, $\frac{dm}{m_0} = \frac{dy}{y_0}$ (in figure)

hence, $dm = \frac{m_0}{y_0} dy$ (3)

Where, $m_0 = \text{mass of the spring,}$

$y_0 = \text{length of the spring,}$

$dm = \text{mass of an element of the spring,}$

$dy = \text{length of an element of the spring,}$

Thus the integral becomes (using equation (2) and (3))

$$\begin{aligned} \int_0^m \frac{1}{2} v^2 dm &= \int_0^{y_0} \frac{1}{2} \left(\frac{y \times v_0}{y_0} \right)^2 \frac{m_0}{y_0} dy = \frac{m_0 v_0^2}{2 y_0^3} \int_0^{y_0} y^2 dy \\ &= \frac{m_0 v_0^2}{2 y_0} \times \frac{y_0^3}{3} = \frac{1}{2} \left(\frac{m_0}{3} \right) v_0^2 \end{aligned}$$

Therefore,

Total kinetic energy (K.E.) of the system will

be = K.E. of the load (suspended mass) +

K.E. of the spring

$$= \frac{1}{2} m v_0^2 + \frac{1}{2} \left(\frac{m_0}{3} \right) v_0^2 = \frac{1}{2} \left(m + \frac{m_0}{3} \right) v_0^2$$

Hence, the effective mass of the vibrating system (load and spring) = $M = m + \frac{m_0}{3}$

Comparing with equation ①, $M = (m + m')$;

hence $m' = \frac{m_0}{3}$; where,

m' = effective mass of the spring,

m_0 = mass of the spring.

So, the effective mass (m') of the spring is one third mass ($m_0/3$) of

the spring.

Spring constant:

The applied force (weight of the load),

$F = mg$ on the spring is proportional to

the extension x of the spring.

within the elastic limit. Therefore

$$F \propto x \quad \text{or} \quad F = kx \quad \text{or} \quad mg = kn \dots (1)$$

where k is spring constant.

Apparatus:

1. A spiral spring
2. Convenient masses with hanging arrangement
3. Top pan balance
4. A rigid framework heavy metal rods.
5. Stop watch
6. ~~me~~ Meterscale

Table: Data for extensions and time periods for different loads

No. of obs.	Load m Kg	Reading of bottom end of the spring on the vertical scale l meter	Extension x meter	Total time for 30 oscillations t/sec	Time period T = (t/30) sec	T ² sec ²
1	0 (No load)	$l_0 = 41.5$	$(l_0 - l_0) = 0$			
2	0.1	$l_1 = 49.3$	$(l_1 - l_0) = 7.8$	19.74	0.658	0.47
3	0.2	$l_2 = 57.2$	$(l_2 - l_0) = 15.7$	27.5	0.9167	0.84
4	0.3	$l_3 = 63.8$	$(l_3 - l_0) = 22.3$	32.44	1.081	1.168
5	0.4	$l_4 = 72.4$	$(l_4 - l_0) = 30.9$	36.39	1.213	1.47
6	0.5	$l_5 = 81$	$(l_5 - l_0) = 39.5$	40.86	1.362	1.855

Calculation:

$$\text{Spring Constant, } K = \frac{mg}{x} = \frac{g}{\frac{x}{m}} = \frac{9.8 \frac{m}{s^2}}{\text{SLOPE of Extension vs Load graph}} = \frac{9.8}{0.78} = 12.5 \text{ N/m}$$

Experimental value of Effective mass (X axis intercept of (Time period)² Vs Load graph), m'

$$m' = 0.048 \text{ Kg.}$$

Mass of the spring measured by weight machine, $m_0 = 0.15 \text{ Kg.}$

$$m' = \frac{m_0}{3} = 0.05 \text{ kg}$$

Standard value (Theoretical value) of Effective mass,

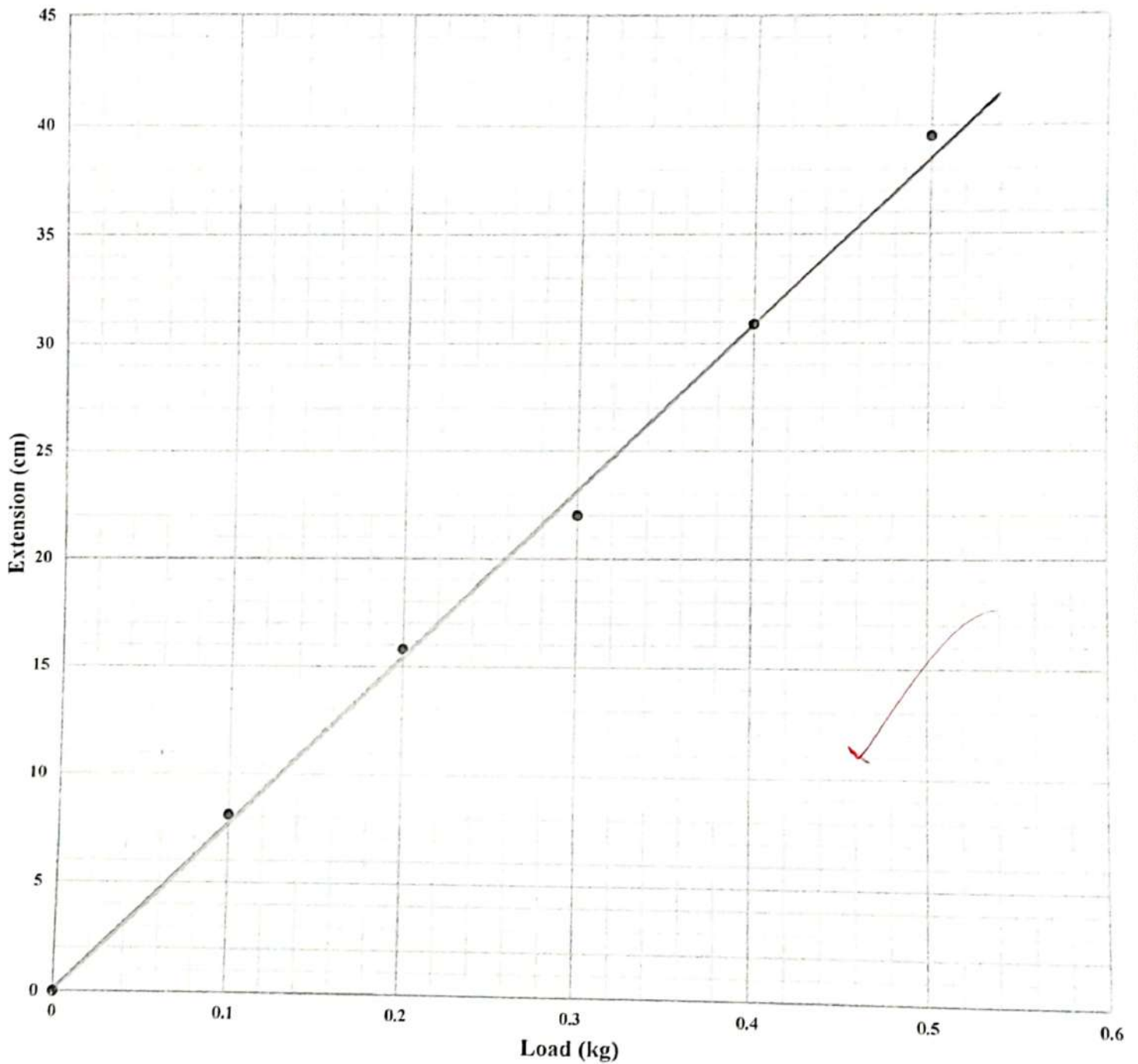
$$\text{Percentage Difference} = \frac{\text{Standard value} - \text{Experimental value}}{\text{Standard value}} \times 100\%$$

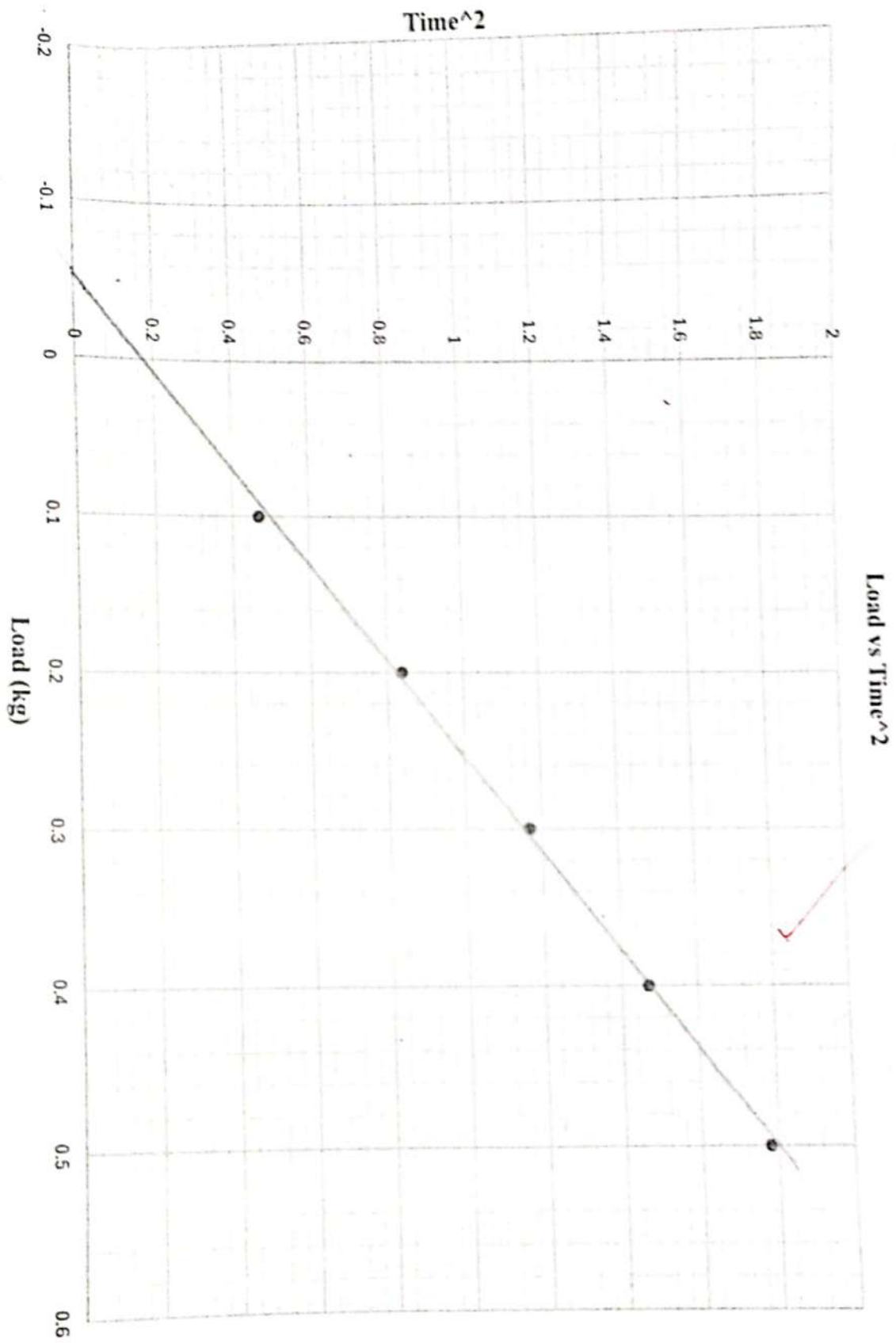
$$\text{Percentage difference} = \frac{0.05 - 0.048}{0.05} \times 100 = 4\%$$

Results:

Spring constant, $k = 12.5 \text{ N/m}$ Effective mass, $m' = 0.048 \text{ kg}$

Load vs Extension





Results:

Spring Constant, $K = 12.5 \text{ N/m}$

Effective mass, $m' = 0.048 \text{ kg}$

Discussions:

1. While hanging the weights that should lie along the axis of spring. This does not occur during the experiment.
2. Oscillation should occur in a vertical plane, but oscillation has occurred little in a horizontal plane as well.
3. Some error in the time period of its oscillation is likely to creep in due to yielding (sometimes) to the support and inaccuracy in the accepted value of mass of load.
4. Should take care that initially the load is pulled only through a small distance before being released gently to let it oscillate vertically.